

Answer food (these answers are the representative answers, there might be other points also)

Q1. Foods have always played the most crucial role in human health. Outline the role of food in human physical, mental and social health.

Answer:

Foods are essential ingredients for our health. The contribution of food to our physical, mental, and social health are describe as bellow

Physical Health:

Nutrient Supply: Foods like fruits, vegetables, and lean proteins provide essential nutrients for growth, tissue repair, and overall body function.

Energy Source: Carbohydrates and fats in foods are the primary energy sources for daily activities.

Protect from diseases: protein, vitamins and minerals protect from different diseases. For example, vitamin A prevents night blindness and vitamin D prevents rickets diseases.

Mental Health:

Brain Function: Omega-3 fatty acids found in fish contribute to cognitive function and memory.

Mood Regulation: Tryptophan obtained from oats, bananas, milk, and tuna fishy supports serotonin production, which influences mood in a positive direction.

Social Health:

Cultural Identity: Traditional foods reflect cultural identity, fostering a sense of belonging and community.

Socialization: Shared meals promote communication, strengthening social bonds with family and friends.

Q2. A 'balanced diet' is more significant than a 'diet' for human health. Differentiate 'diet' and 'balanced diet' and outline the characteristics of a 'balanced diet'.

Ans:

Diet: The amount of food that a person consumes in a day is called the diet of the person for the day

Balanced Diet: *Balanced diet is that food in which all the nutritious elements are present in proper quantity and proportion according to the physical demand, age, weight, activity, etc.*

Characteristics of Balanced Diet (any four)

1. The diet should be sufficient to provide energy
2. Balanced diet should contain all the nutrients in proper proportion.
3. The number of nutrients stored in the body should be high in the diet.

4. A balanced diet should include foods from all the food groups.
5. Food should be attractive, organic/natural, and tasty.
6. A balanced diet typically contains 50-60% carbohydrates, 12-20% protein, and 30% fat

Q3. Although 'nutrition' and 'nutrients' sound similar in pronunciation, they serve different functions. Differentiate 'nutrition' and 'nutrients' with example.

Answer:

Nutrition (characteristics)	Nutrients (characteristics)
a) Nutrition is a combination of processes through which food intake and utilization of nutrients obtained from the food take place.	a) Nutrients are components present in food that an organism can use to survive and grow.
b) Through the process of nutrition, nutrients produced from the taken food	b) Nutrients are obtained from the food
c) Nutrition includes food intake, digestion, absorption, assimilation and excretion of waste.	c) The major nutrients in our food are carbohydrates, fats, proteins, vitamins and minerals.

Q4. Complete the table utilizing the concept of food groups

Food	Origin	Nutrients	Function
Rice	Plant	Carbohydrates	Energy source
Milk	Animal	Protein, Fat	Energy source, Body building, Protection from diseases
Soybean	Plant	Protein	Energy source, Body building, Protection from diseases
Green leafy vegetable	Plant	Vitamins, Minerals	Body building, Protection from diseases
Paneer	Animal	Protein, Fat	Energy source, Body building, Protection from diseases
Rock Salt	Mineral	Mineral	Body building, Protection from diseases

OR

Vitamins play a crucial role in human health and growth. Illustrate the sources of fat-soluble vitamins and explore the impact of deficiency of these vitamins on human health.

Ans: Fat soluble vitamin (any two sources and any two impacts of each)

Vitamins	Sources	Impact of deficiency
Vitamin K	green leafy vegetables (e.g., kale, spinach, broccoli); fermented foods (e.g., natto, cheese), animal products (Fish, liver,)	Impaired blood clotting, excessive bleeding, easy bruising, bone health issues
Vitamin D	Sunlight exposure, fatty fish, egg yolks, fortified foods (e.g., milk, cereal)	Rickets, osteomalacia, muscle weakness, immune dysfunction
Vitamin A	Preformed vitamin A: liver, fish, dairy products; Provitamin A carotenoids: carrots, sweet potatoes, spinach, kale	Night blindness, impaired immune function, xerophthalmia, skin disorders
Vitamin E	Plant-based oils (e.g., wheat germ oil, sunflower oil, olive oil), nuts, seeds, green leafy vegetables	Muscle weakness, vision problems, neurological abnormalities, reproduction issue, aging issue

Q5. Let us consider Miss. Lopez is a moderate worker. She needs 600 calories for her daily work apart from BMR. The details of Miss. Lopez's height, weight and age are given in the table. Calculate the total calories required for Miss Lopez per day.

Height (cm)	Weight (kg)	Age (year)
150	48	35

Ans:

Total energy =

BMR + working energy

$$\text{BMR} = 447.6 + (9.25 \times 48) + (3.1 \times 150) - (4.33 \times 35) \text{ cal/day}$$

$$\approx 1205.05$$

Or

$$\text{BMR} = 447.593 + (9.247 \times 48) + (3.098 \times 150) - (4.33 \times 35) \text{ cal/day}$$

$$\approx 1204.599 \text{ cal/day}$$

Total energy

$600 + 1205 = 1805$ cal/day (around this)

Total energy required for the lady is 1805 cal/day

Q6. Proper nutrition plays a vital role in human growth and development. Differentiate optimum, under and overnutrition, and illustrate their impact on human health.

Ans:

- **Optimum nutrition:** Optimum nutrition is also called best nutrition. It can be defined as taking the right amount of nutrients on proper schedule to achieve the best performance and the longest lifetime in good health.

Impact on body

1. The body of a person receiving optimum nutrition remains strong and there is a balance in his body length proportion.

2. The body's digestive organs, blood circulation, and respiratory system function well. The muscles of a well-nourished person are healthy and strong.

- **Under Nutrition:** Under nutrition is a condition that results from eating diets for long time in which one or more nutrients are not enough and this diet causes health problems. This health condition called under nutrition

Impact on body

Impact of under nutrition on body

Wasting: Low weight with respect to height.

Stunting: Shorter height with respect to age

Underweight: Low weight with respect to age

- **Over nutrition:** Over nutrition is a condition that results from eating diets for long time in which one or more nutrients are in excess and this diet causes health problems. This health condition called under nutrition

Impact on human body**Obesity:** A disorder which occurs by eating more calories than one burns

Oversupplying a specific nutrient, such as dietary, minerals, or vitamin poisoning.

For mineral excess: Iron poisoning (vomiting, Liver damage)

Diabetes Hypertension, Cardiovascular disease, and stroke

Q7. Minerals are essential for the proper growth and development of humans. Illustrate the function of calcium and phosphorus on human health.

Ans: Calcium (Ca)

Functions (any three)

1. **Bone and Teeth Health:** Calcium is a major component of bones and teeth, providing structural support and strength. It helps to maintain bone density and prevents conditions like osteoporosis, which is characterized by weak and brittle bones.
2. **Muscle Contraction:** Calcium ions are essential for muscle contraction. When a muscle is stimulated, calcium is released from storage sites within the muscle cells, allowing the muscle fibers to contract. This process is fundamental for movement, including activities such as walking, running, and lifting.
3. **Nerve Function:** Calcium is involved in transmitting nerve impulses throughout the nervous system. It helps regulate the release of neurotransmitters (acetylcholine, ions), which are chemical messengers that allow nerve cells to communicate with each other. Proper calcium levels are necessary for the proper functioning of the nervous system.
4. **. Blood Clotting:** Calcium plays a vital role in the blood clotting process, also known as coagulation. When a blood vessel is injured, platelets in the blood release calcium ions, which trigger a series of reactions leading to the formation of a blood clot. This mechanism helps to stop bleeding and promote wound healing.
5. **Cell Signaling:** Calcium ions act as second messengers in many cellular signaling pathways. They regulate various cellular processes such as cell growth, proliferation, and apoptosis (programmed cell death). Calcium signaling is involved in controlling gene expression, enzyme activity, and hormone secretion.
6. **Enzyme Activation:** Calcium ions serve as cofactors for several enzymes, meaning they are required for the enzymes to function properly. These enzymes are involved in numerous metabolic reactions within cells, including energy production, DNA synthesis, and protein metabolism.

Phosphorous (P)

Functions (any three)

1. **Auxiliary in the formation of bones and teeth-**Phosphorus combines with calcium to form calcium phosphate (insoluble salts) which make bones and teeth strong. Calcium phosphorous remains in 1:5 ratio in bones. **It is essential for skeletal growth and general health.**

2. **Formation of Nucleoproteins and Nucleic acids**-Phosphorus is an essential substance for the formation of nucleoproteins and nucleic acids. Nucleoprotein resides in the cytoplasm and nucleus of our cells and here this cell helps in reproduction and exchange of hereditary properties.
3. **Maintaining the balance of acid-base in the body**—The specialty of phosphorus is that it can combine with more hydrogen ion. Due to this characteristic it prevents the body from being acidic and maintains the balance of acid base in the blood. The amount of phosphorus in the blood serum of adults is 2.5-4 mg / 100 ml and the amount of phosphorus in the serum of children is 4.5 mg / 100 ml.
4. **Helps in the energy production process**-Phosphorus is an essential component when the liver reuses the glycogen stored in the muscles to get energy, because at that time this glycogen is in the form of a phosphorus-containing glucose compound.
5. **Helps in Carbohydrate Metabolism**-Glucose in the blood is taken up into your body's cells and used to produce a fuel molecule called adenosine triphosphate (ATP) through a series of complex processes known as cellular respiration. Cells can then use ATP to power a variety of metabolic tasks.. Phosphorus combines with other elements to form ATP.
6. **Facilitates the transport of nutrients**-Fats are transported in the blood in the form of phospholipids.

Q8. Imagine you're advising someone about his/her diet and you know that too much and too low of nutrients are bad for the health. Outline your explanation about the impact of deficiency and excess of protein on human health.

Ans:

Deficiency of protein (any three)

- 1) **Kwashiorkor**-Kwashiorkor was discovered by Sisley William in 1935. It means “the disease that occurs to the first child after the birth of the second child.” The following symptoms are seen in children suffering from Kwashiorkor:

Marasmus—This disease occurs due to a deficiency of both protein and energy. After the age of 6 months, apart from the mother’s milk, other food items are also necessary for the children, but if the children do not get them, there is a deficiency in the nutritional elements.
- (2) **Wasting of Muscles**—The muscles of the body start getting destroyed. Its effect is more on the muscles of the arms, hands, and legs. The arms, hands, and legs become thin and weak.
- (3) **Mental changes**-Apathy and irritability come in the nature of the child. Children start crying while talking and they do not want to play the child feels sluggish, lethargic, and tired.
- (4) **Anemia**-Due to lack of protein, hemoglobin is not formed. Hence. there is a disease of red blood cell deficiency.

(5) Effect of fat in the liver-Due to the lack of protein, fat deposits on the liver, due to which its size increases. The size of the pancreas becomes smaller.

7) Effects on the skin-Due to lack of protein, the skin becomes dry and dull. There is a change in the colour of the skin. There are dark brown and black marks on the skin.

(8) Effect on hair-Hair becomes dry, rough, and dull. There is a change in the colour of the hair. Hair turns gray or white, and hair growth stops.

(9) Effects on Mucous Membrane-Mucous membranes are affected due to lack of protein. Due to this, the lips begin to crack and the surface of the tongue becomes greasy.

(10) Moon-shaped Face—In addition to the hands and feet being swollen, the swelling also occurs on the face, due to which the child's face becomes round like the moon.

(12) Effect on Nervous System-Due to lack of protein, not only the physical development of the child stops, but mental development also stops. The child's ability to learn decreases. Endocrine hormones are not produced and secreted properly. There is a disturbance in blood circulation.

Excess of protein (any three)

1. **Kidney Strain:** Excessive protein intake can put a strain on the kidneys as they are responsible for filtering and eliminating waste products from protein metabolism, including nitrogen. Over time, this increased workload can lead to kidney damage or dysfunction.
2. **Dehydration:** High-protein diets often result in increased water loss through urine. This can lead to dehydration if adequate fluid intake is not maintained.
3. **Mineral Imbalances:** High protein intake can disrupt the balance of minerals in the body, particularly calcium. It may lead to increased calcium excretion through urine, potentially contributing to a **risk of osteoporosis and kidney stones**.
4. **Digestive Issues:** Excess protein can be hard on the digestive system, potentially causing constipation, bloating, and other gastrointestinal discomfort.
5. **5. Increased Risk of Heart Disease:** Some high-protein diets, particularly those high in animal protein, may also be high in saturated fats and cholesterol, increasing the risk of heart disease.
6. **6.Weight Gain:** Contrary to some beliefs, excessive protein consumption alone may not necessarily lead to weight loss. If the excess protein intake is not balanced with other nutrients and calorie control, it can contribute to weight gain.
7. **7. Bad Breath and Body Odor:** Excess protein metabolism can produce ammonia, which can lead to bad breath and unpleasant body odor.

8. **8. Liver Disorders:** In rare cases, excessive protein intake can stress the liver and potentially lead to liver disorders.
9. **9. Nutrient Imbalances:** Overemphasis on protein intake can sometimes lead to a lack of other essential nutrients, as individuals may prioritize protein-rich foods over a balanced diet containing fruits, vegetables, and whole grains.

OR

Imagine you're discussing dietary choices with a group of individuals and they are curious about the role of fats in their bodies. Outline the detailed overview of the function of fats in the human body.

Ans: Fat performs the following functions in the body (any six)

- 1. Providing energy**—A lot of energy is released from fat. Usually 9 calories of energy is released from one gram of fat.
- 2. Storage of energy**—The amount of excess fat in the body is present in the form of adipose tissue below the skin in the form of a layer. At the time of need, this stored fat is used for energy production and when energy is not needed then this fat keeps on accumulating in the fatty fibres, due to which the person becomes fat.
- 3. Regulating body temperature**—A normal layer of fat remains under the skin which acts as a bad conductor of heat in the body. Being an insulator, the temperature of the body remains regulated by the environmental influence.
- 4. Obtaining fat soluble vitamins**—Fat is necessary for the absorption of vitamins A, D, E, K. If the amount of fat in the diet is low, these vitamins will not be able to accumulate, and various deficiency diseases will occur.
- 5. Providing essential fatty acids**—Some essential fatty acids are not synthesized by the body, it is necessary to supply them through food. The deficiency of essential fatty acids synthesis leads to various skin diseases. It is necessary to take them in good quantity to keep the body healthy.
- 6. To protect the soft parts of the body**—Some essential fatty acids protect soft organs like the kidney, the heart and nervous system from external shocks by forming a deep structure. The highly sensitive organs are also covered with a double layer of fat to protect the soft organs from external shocks.
- Prolonged satisfaction**—Fat slows down the secretion of digestive juices. If the digestive juices are secreted late, then the process of digestion will take time. Therefore, food will remain in the digestive organs for a long time, due to which the person feels the satisfaction of food for a longer time and does not feel hungry.
- 8. Saving protein**—If energy is not supplied by carbohydrates, then protein also does the work of energy production leaving its main building work. In such a situation, if the amount of fat is present in the body, the fat prevents the protein from being used for energy production, thus saving the protein.